

Substitute into formulae

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$. Copy or complete the first step.

2. Find the first five terms of $4n - 1$.

3. Continue 7, 12, 17, 22 and state the rule.

4. Miro says: Miro treats a variable as a label instead of a number that can vary. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$. Copy or complete the first step.
Answer 31.
2. Find the first five terms of $4n - 1$.
Answer 3, 7, 11, 15, 19.
3. Continue 7, 12, 17, 22 and state the rule.
Answer 27, 32; add 5.
4. Miro says: Miro treats a variable as a label instead of a number that can vary. Is this correct? Explain.
Answer State what the variable represents, substitute its value and preserve the operation structure.

Substitute into formulae

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.

6. Check this result using an inverse, estimate or known fact: Continue 7, 12, 17, 22 and state the rule.

2. Find the first five terms of $4n - 1$.

7. Prove or explain your reasoning: Find the missing terms: 84, __, 62, __, 40.

3. Continue 7, 12, 17, 22 and state the rule.

8. Identify and correct this misconception: Miro treats a variable as a label instead of a number that can vary.

4. Solve this independently and show every stage: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Find the first five terms of $4n - 1$.

10. Write and solve a similar question to: Continue 7, 12, 17, 22 and state the rule.

Teacher answers and checking prompts

1. Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.
Answer 31.
2. Find the first five terms of $4n - 1$.
Answer 3, 7, 11, 15, 19.
3. Continue 7, 12, 17, 22 and state the rule.
Answer 27, 32; add 5.
4. Solve this independently and show every stage: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.
Answer 31.
5. Use a second method or representation for: Find the first five terms of $4n - 1$.
Answer Expected result: 3, 7, 11, 15, 19. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Continue 7, 12, 17, 22 and state the rule.
Answer Expected result: 27, 32; add 5. A valid check must be shown.
7. Prove or explain your reasoning: Find the missing terms: 84, __, 62, __, 40.
Answer 73 and 51. The explanation must show why the method works.
8. Identify and correct this misconception: Miro treats a variable as a label instead of a number that can vary.
Answer State what the variable represents, substitute its value and preserve the operation structure.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Continue 7, 12, 17, 22 and state the rule.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Substitute into formulae

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Find the missing terms: 84, __, 62, __, 40.

6. Create a new example that tests the same idea as: Continue 7, 12, 17, 22 and state the rule.

2. Prove the answer to this question, rather than only calculating it: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Find the first five terms of $4n - 1$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 27, 32; add 5.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro treats a variable as a label instead of a number that can vary.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find the missing terms: 84, __, 62, __, 40.
Answer 73 and 51. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.
Answer Expected result: 31. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Find the first five terms of $4n - 1$.
Answer Expected result: 3, 7, 11, 15, 19. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 27, 32; add 5.
Answer One valid reconstruction is: Continue 7, 12, 17, 22 and state the rule. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro treats a variable as a label instead of a number that can vary.
Answer State what the variable represents, substitute its value and preserve the operation structure.
6. Create a new example that tests the same idea as: Continue 7, 12, 17, 22 and state the rule.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.